

ASAD HUSSAIN

asadh1521@gmail.com · +92 320 4141092 · Lahore, Pakistan · [linkedin.com/in/asad-hussain92](https://www.linkedin.com/in/asad-hussain92)

PROFESSIONAL SUMMARY

Final-year Electrical Engineering student at NUST working across applied machine learning and embedded/RF systems engineering. Built content-scoring backend systems at FlyRank AI and trained in deep learning, NLP, LLMs, and RAG through AtomCamp's AI Bootcamp; on the hardware side, designed DroneGuard, a two-board RF detection and jamming system, alongside embedded firmware and IoT telemetry from prior internships. Comfortable moving between training models and building the circuits and firmware that feed them real-world data.

TECHNICAL SKILLS

Programming: Python · C++/Embedded C · JavaScript · SQL · MATLAB/Simulink · Flask

ML / AI: Scikit-learn · TensorFlow · PyTorch · LangChain · RAG · LLM Fine-tuning · FastAPI · Docker

Hardware & RF: SDR · STM32 · Arduino · Raspberry Pi · PCB Design (Proteus) · Antenna Design

Tools: Git · REST APIs · SQL Server · Google Cloud · Weights & Biases · AutoCAD

Signal Processing: FFT / Spectrogram Features · RF Signal Classification · MATLAB ML Toolbox · Dataset Labelling

WORK EXPERIENCE

Technical Intern | National Electric Power Regulatory Authority (NEPRA), Islamabad

Jul – Aug 2026

- Studied NEPRA's end-to-end regulatory flow (license → tariff → sale) across Generation, Transmission, Distribution, and Supply licenses, including the three generation tariff mechanisms (cost-plus, upfront, competitive bidding).
- Studied AMI meter architecture and PLC/cellular communication protocols, including recurring non-communicating meter issues flagged across DISCOs.
- Analyzed NGC's transmission network data — 233 grid-station transformers (116 above 80% rated capacity, annual T&T losses of 2.569% against a 2.50% threshold) — and NEPRA's State of the Industry Report, covering 111,466 GWh of national consumption.

Engineering Intern, Full-Stack Systems | Research & Indigenous Development Centre (RDC), HIT

Jun – Aug 2025

- Integrated and wired core hardware modules including the LilyGO T-SIM7670E (ESP32 core + LTE modem), LilyGO T3-S3 GPS module, and RDM6300 RFID reader with a TP4056 LiPo charge controller.
- Programmed low-level control logic and state machines in C/C++ for 1Hz GPS data capture, RFID card authentication, and buzzer-based audio feedback.
- Implemented MQTT messaging over cellular GPRS with QoS Level 1, enabling reliable real-time telemetry with 50ms server-side callback processing.
- Built an asynchronous Python Flask backend with multithreading to process analytics workloads without blocking incoming MQTT streams.

FINAL YEAR PROJECT

DroneGuard — Anti-Drone Surveillance System

2025 – 2026

SDR · GNU Radio · Raspberry Pi 4 · STM32H743 · MATLAB · Python

- Designed a two-board anti-drone system: a four-band RF detection front-end (433 MHz–2.4 GHz) and a jamming board built around an ADF4351 PLL synthesiser.
- Built a real-time signal-classification pipeline (RTL-SDR + GNU Radio) identifying FHSS, OFDM, and FSK drone protocols from live IQ samples, plus a MATLAB front-end for live spectrum and telemetry.
- Integrated a low-noise amplifier (+18.7 dB gain, 0.6 dB noise figure) that extended detection range from ~30 m to 150–200 m.
- Authored a full component-level technical reference and costed the full system at roughly a 98% reduction versus commercial anti-drone platforms.

EDUCATION

National University of Sciences and Technology (NUST)

2023 – 2027

B.E. Electrical Engineering — Coursework: Analog Electronics, DSP, Control Systems, Communication Systems, Digital Logic Design, Embedded Systems, Signals & Systems, Instrumentation, OOP, Microprocessor Systems

AI/ML TRAINING — ATOMCAMP

Completed AtomCamp's AI Bootcamp, a Pakistan-founded AI & data science training program, covering machine learning & deep learning (regression, SVM, CNNs, transfer learning, YOLO), NLP & LLMs (embeddings, transformers, fine-tuning), generative AI & RAG (LangChain agent development), and MLOps & deployment (Docker, FastAPI, CI/CD, Google Cloud).

ACADEMIC PROJECTS

Hands-on projects across core EE disciplines: Butterworth Active LPF (op-amp) · Multi-Stage Audio Amplifier · EEG Cleaning MATLAB GUI · PID DC Motor Controller (Arduino) · 5-Band Graphic Equaliser (Simulink) · 4-Bit ALU · FM Receiver (TDA7000) · Multi-Range Voltmeter · OOP Shopping Cart (C++) · Water Level Indicator (Proteus) · Current Transformer

CERTIFICATIONS, ACTIVITIES & ACHIEVEMENTS

AtomCamp AI Bootcamp · MATLAB Fundamentals (MathWorks) · PCB Design (Proteus & SPICE) · Technical Lead, NUST Robotics Society · Highest Achiever, Matriculation (SSC) · 1st Place Robotics, Voltfest · NMUN 2024 Delegate

PORTFOLIO

[Asad Hussain | Electrical Engineer & IoT Developer](#) — full-stack portfolio showcasing embedded, RF/SDR, and ML/AI projects.